



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.



Dataset at a Glance

3,900

Transactions

Total purchase records analyzed

18

Features

Demographics, purchase details, and behavior

37

Missing Values

In Review Rating column only

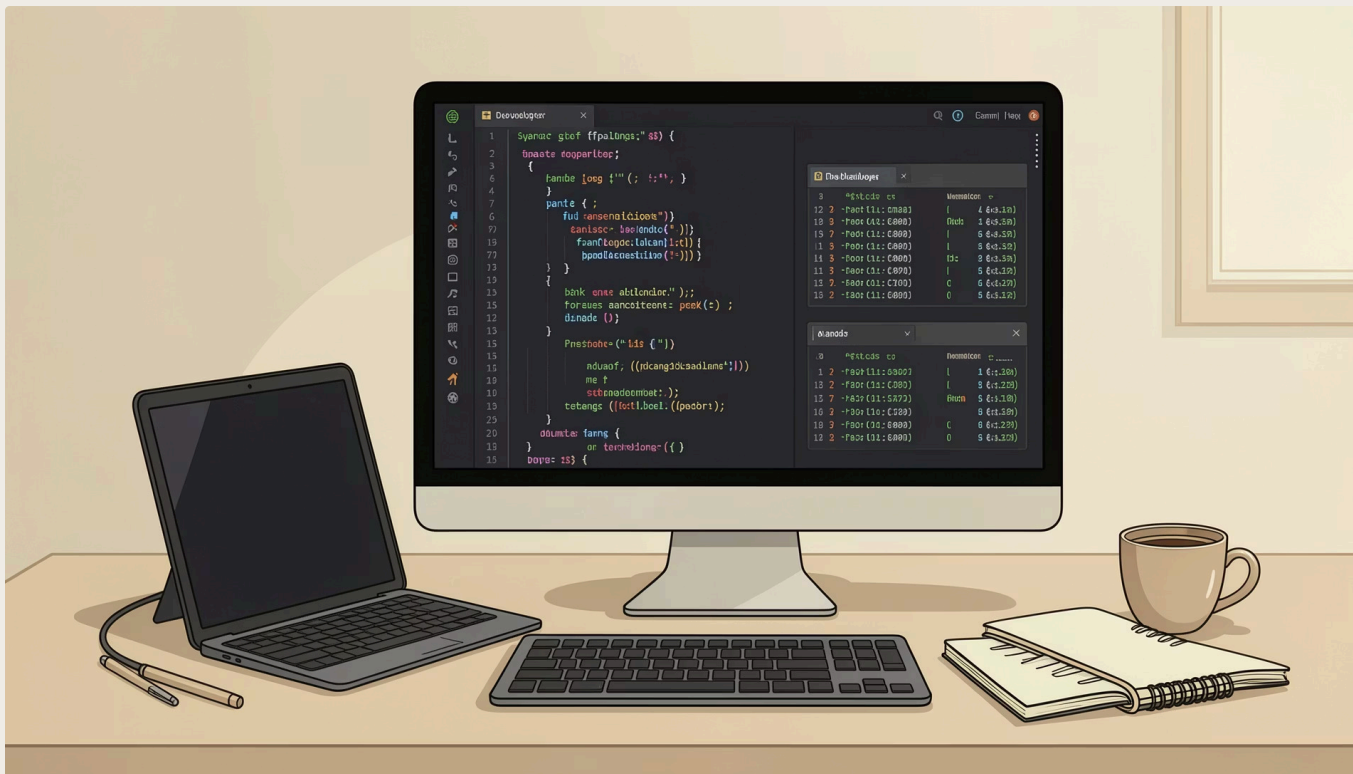
\$59.76

Avg. Purchase

Average transaction amount (USD)

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount Applied, Frequency, Review Rating, Shipping Type).

Data Preparation in Python



→ Load & Explore

Imported dataset via `pandas`; used `df.info()` and `df.describe()` for initial exploration.

→ Clean & Standardize

Imputed missing Review Ratings using median per category; renamed columns to `snake_case`.

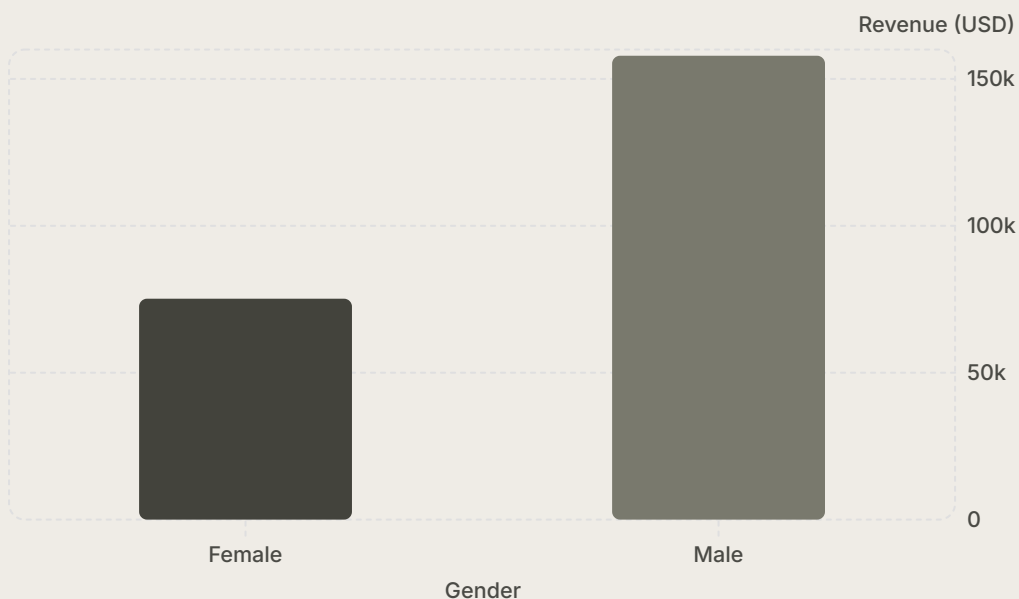
→ Feature Engineering

Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`.

→ Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.

Revenue by Gender & Shipping Type



Key Findings

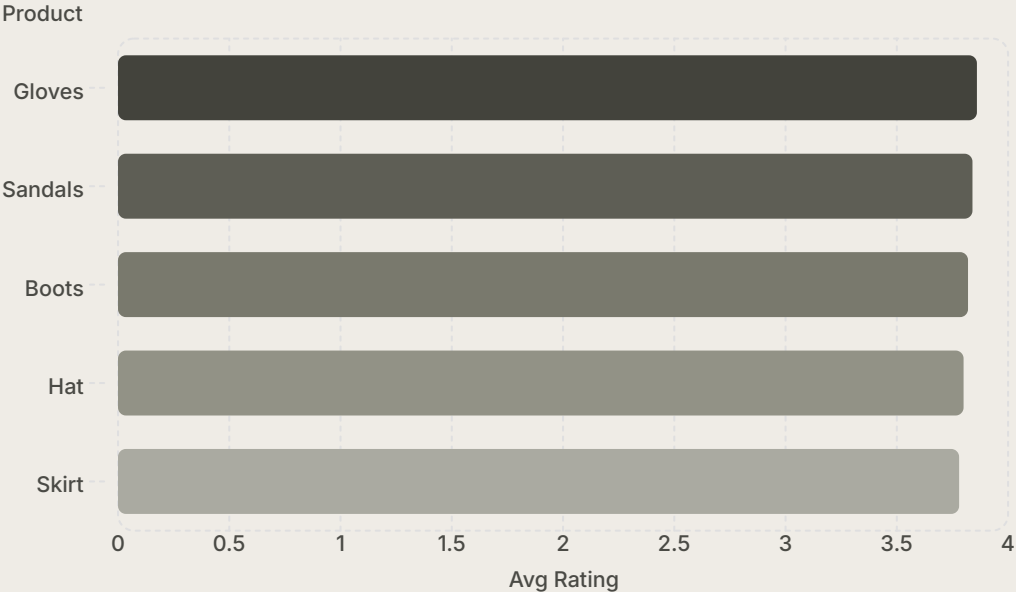
Male customers generated **2.1× more revenue** than female customers (\$157,890 vs \$75,191).

Express shipping customers spent slightly more on average (\$60.48) vs Standard shipping (\$58.46).

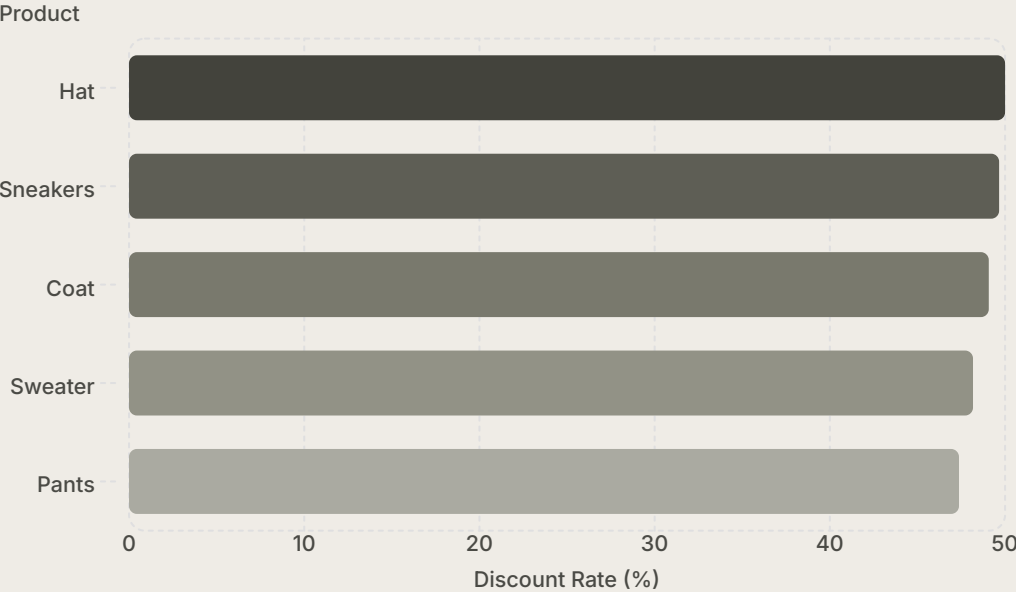
i 839 customers used discounts yet still spent above the average purchase amount.

Top Products by Rating & Discount Rate

Highest Rated Products



Most Discounted Products



Customer Segmentation

Loyal

3,116 customers — The dominant segment with the highest purchase frequency.

Returning

701 customers — Repeat buyers with growth potential toward loyalty.

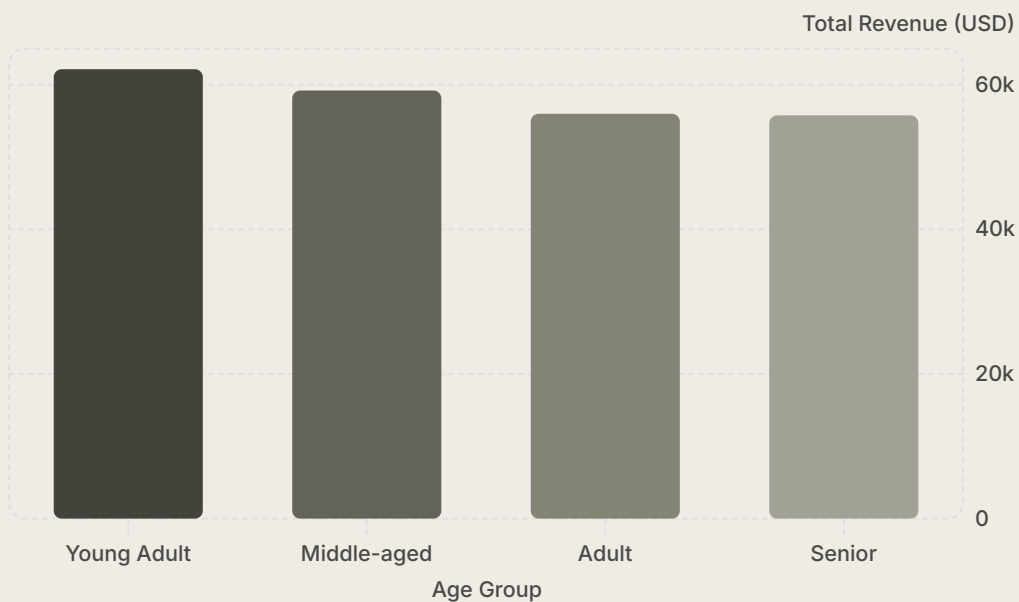
New

83 customers — First-time buyers; prime targets for onboarding campaigns.

Of customers with more than 5 purchases, **958 are subscribers** vs 2,518 non-subscribers — indicating room to convert repeat buyers.



Subscription Behavior & Age Group Revenue

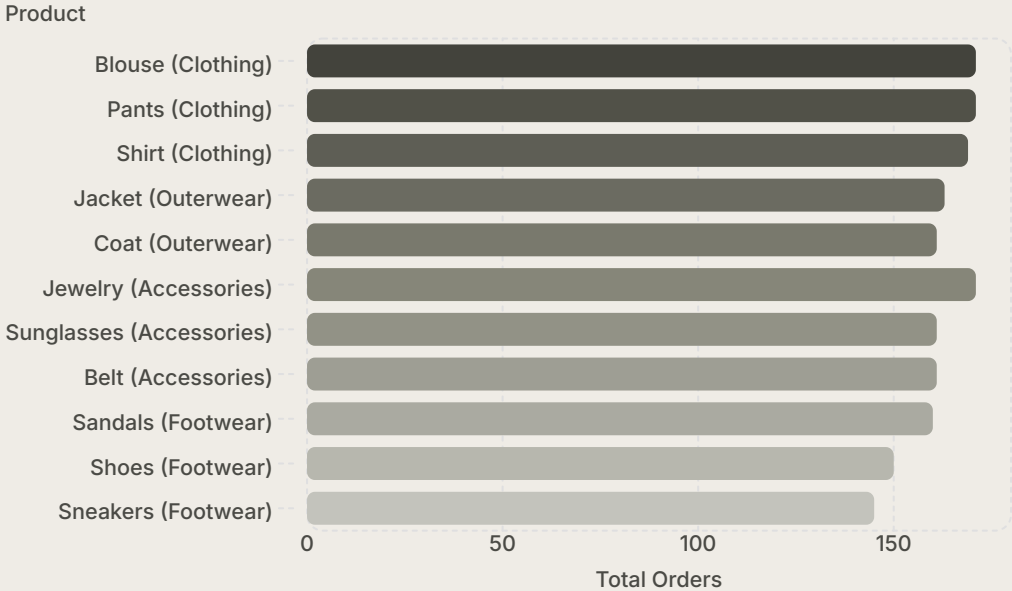


Subscription Breakdown

Only **27% of customers** are subscribers (1,053), yet they represent a high-value segment worth nurturing.

- Subscribers avg spend: **\$59.49**
- Non-subscribers avg spend: **\$59.87**
- Young Adults lead revenue at **\$62,143**

Top 3 Products per Category



Category Leaders

Clothing dominates with Blouse, Pants, and Shirt as top sellers.
Accessories led by Jewelry. Footwear topped by Sandals.
Outerwear led by Jacket and Coat.

Power BI Dashboard

An interactive dashboard was built to present all insights visually, featuring KPIs, revenue breakdowns, and customer segment filters.

3.9K Customers

Total transactions tracked

\$59.76 Avg

Average purchase amount

3.75 Rating

Average review score

27% Subscribers

Subscription penetration

Business Recommendations



Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscriber base.



Customer Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.



Review Discount Policy

Balance sales boosts with margin control on heavily discounted items.



Targeted Marketing

Focus on high-revenue age groups (Young Adults) and express-shipping users.



Product Positioning

Highlight top-rated and best-selling products in campaigns.